Bregman Divergences, Csiszár Divergences, and the Uniqueness of KL

1 General measure-theoretic setting

Let $(\mathcal{X}, \mathcal{A})$ be a measurable space. We write α and β for probability measures on $(\mathcal{X}, \mathcal{A})$.

For the Bregman part, it is convenient to work on a dominated model. Thus fix a finite reference measure λ on $(\mathcal{X}, \mathcal{A})$, and consider probability measures of the form

$$\alpha = p\lambda, \qquad \beta = q\lambda,$$

where

$$p = \frac{\mathrm{d}\alpha}{\mathrm{d}\lambda}, \qquad q = \frac{\mathrm{d}\beta}{\mathrm{d}\lambda}.$$

To avoid boundary issues, define the positive bounded part of the model by

$$\mathcal{P}_{\lambda,*} = \left\{ p\lambda : p \in L^\infty(\lambda),\ \int_{\mathcal{X}} p \,\mathrm{d}\lambda = 1,\ \text{and there exist } 0 < m < M < \infty \text{ such that } m \le p \le M \right\}.$$

The tangent directions are signed densities $h \in L^\infty(\lambda)$ satisfying

$$\int_{\mathcal{X}} h \,\mathrm{d}\lambda = 0.$$

If $p \in \mathcal{P}_{\lambda,*}$ and h is such a tangent direction, then $p + \varepsilon h$ is again positive for all sufficiently small ε.

2 Bregman divergence on a space of measures

Let Φ be a convex functional on a convex class of probability measures. If Φ is differentiable at β in the signed direction $\alpha - \beta$, its Bregman divergence is

$$\mathrm{B}_\Phi(\alpha \mid \beta) = \Phi(\alpha) - \Phi(\beta) - D\Phi(\beta)[\alpha - \beta].$$

In the dominated notation $\alpha = p\lambda$, $\beta = q\lambda$, this is

$$\mathrm{B}_\Phi(p\lambda \mid q\lambda) = \Phi(p) - \Phi(q) - D\Phi(q)[p - q].$$

Here

$$D\Phi(q)[h] = \frac{\mathrm{d}}{\mathrm{d}\varepsilon} \Phi(q + \varepsilon h) \Big|_{\varepsilon = 0}$$

is the first variation of Φ at q in the direction h.

By convexity,

$$\mathrm{B}_\Phi(\alpha \mid \beta) \ge 0.$$

Also, adding an affine functional to Φ does not change B_Φ.

3 Csiszár, or f-divergence

Let

$$\varphi : (0, \infty) \to \mathbb{R}$$

be convex and satisfy

$$\varphi(1) = 0.$$

If $\alpha \ll \beta$, the Csiszár divergence generated by φ is

$$\mathrm{D}_\varphi(\alpha \mid \beta) = \int_{\mathcal{X}} \varphi\left(\frac{\mathrm{d}\alpha}{\mathrm{d}\beta}\right) \mathrm{d}\beta.$$

In dominated form, if $\alpha = p\lambda$ and $\beta = q\lambda$ with $q > 0$, then

$$\frac{\mathrm{d}\alpha}{\mathrm{d}\beta} = \frac{p}{q},$$

and therefore

$$\mathrm{D}_\varphi(\alpha \mid \beta) = \int_{\mathcal{X}} q(x)\, \varphi\left(\frac{p(x)}{q(x)}\right) \mathrm{d}\lambda(x).$$

For arbitrary probability measures α, β, one can use the usual Lebesgue decomposition

$$\alpha = \alpha_{\mathrm{ac}} + \alpha_{\mathrm{s}}, \qquad \alpha_{\mathrm{ac}} \ll \beta, \qquad \alpha_{\mathrm{s}} \perp \beta.$$

Writing

$$r = \frac{\mathrm{d}\alpha_{\mathrm{ac}}}{\mathrm{d}\beta}$$

and

$$\varphi'_\infty = \lim_{t \to \infty} \frac{\varphi(t)}{t},$$

the extended Csiszár divergence is

$$\mathrm{D}_\varphi(\alpha \mid \beta) = \int_{\mathcal{X}} \varphi(r)\, \mathrm{d}\beta + \varphi'_\infty\, \alpha_{\mathrm{s}}(\mathcal{X}).$$

In the positive dominated model $\mathcal{P}_{\lambda,*}$, the singular term is absent.

4 The KL example

Define the entropy functional relative to λ by

$$H_\lambda(p) = \int_{\mathcal{X}} p \log p \, \mathrm{d}\lambda.$$

Its first variation is

$$DH_\lambda(q)[h] = \int_{\mathcal{X}} (1 + \log q) h \, \mathrm{d}\lambda.$$

Therefore

$$\begin{aligned}
\mathrm{B}_{H_\lambda}(p\lambda \mid q\lambda) &= \int_{\mathcal{X}} p \log p \, \mathrm{d}\lambda - \int_{\mathcal{X}} q \log q \, \mathrm{d}\lambda - \int_{\mathcal{X}} (1 + \log q)(p - q)\, \mathrm{d}\lambda \\
&= \int_{\mathcal{X}} p \log p \, \mathrm{d}\lambda - \int_{\mathcal{X}} p \log q \, \mathrm{d}\lambda - \int_{\mathcal{X}} p \, \mathrm{d}\lambda + \int_{\mathcal{X}} q \, \mathrm{d}\lambda \\
&= \int_{\mathcal{X}} p \log \frac{p}{q} \, \mathrm{d}\lambda.
\end{aligned}$$

Since $p = \mathrm{d}\alpha/\mathrm{d}\lambda$ and $q = \mathrm{d}\beta/\mathrm{d}\lambda$, this is

$$\mathrm{B}_{H_\lambda}(\alpha \mid \beta) = \int_{\mathcal{X}} \log\left(\frac{\mathrm{d}\alpha}{\mathrm{d}\beta}\right) \mathrm{d}\alpha = \mathrm{KL}(\alpha \mid \beta).$$

On the other hand, if

$$\varphi(t) = t \log t,$$

then

$$\mathrm{D}_\varphi(\alpha \mid \beta) = \int_{\mathcal{X}} q \left(\frac{p}{q} \log \frac{p}{q}\right) \mathrm{d}\lambda = \int_{\mathcal{X}} p \log \frac{p}{q} \, \mathrm{d}\lambda = \mathrm{KL}(\alpha \mid \beta).$$

Thus KL is both a Bregman divergence and a Csiszár divergence.

5 Uniqueness theorem

Theorem 1. *Assume that $(\mathcal{X}, \mathcal{A}, \lambda)$ is nontrivial, in the sense that there is a measurable set $A \in \mathcal{A}$ with*

$$0 < \lambda(A) < \lambda(\mathcal{X}) < \infty.$$

Let Φ be a convex functional on $\mathcal{P}_{\lambda,*}$, twice Gateaux differentiable along bounded zero-mass directions. Let

$$\varphi \in C^2(0, \infty)$$

be convex and satisfy $\varphi(1) = 0$.

Suppose that

$$\mathrm{B}_\Phi(\alpha \mid \beta) = \mathrm{D}_\varphi(\alpha \mid \beta)$$

for all $\alpha, \beta \in \mathcal{P}_{\lambda,*}$. Then there exist constants $c \geq 0$ and $a \in \mathbb{R}$ such that

$$\varphi(t) = c\, t \log t + a(t - 1).$$

Consequently,

$$\mathrm{B}_\Phi(\alpha \mid \beta) = \mathrm{D}_\varphi(\alpha \mid \beta) = c \ \mathrm{KL}(\alpha \mid \beta).$$

Moreover,

$$\Phi = c\, H_\lambda + \ell$$

on $\mathcal{P}_{\lambda,*}$, where ℓ is affine. Hence, apart from the trivial zero divergence and scalar multiplication, the only common divergence is KL.

Proof. Fix $\beta = q\lambda \in \mathcal{P}_{\lambda,*}$ and let $h \in L^\infty(\lambda)$ satisfy

$$\int_{\mathcal{X}} h \, \mathrm{d}\lambda = 0.$$

For sufficiently small ε, define

$$\alpha_\varepsilon = (q + \varepsilon h)\lambda.$$

Then $\alpha_\varepsilon \in \mathcal{P}_{\lambda,*}$.

By the assumed equality of the two divergences,

$$\mathrm{B}_\Phi(\alpha_\varepsilon \mid \beta) = \mathrm{D}_\varphi(\alpha_\varepsilon \mid \beta).$$

First compute the second variation of the Bregman side. Since

$$\mathrm{B}_\Phi(\alpha_\varepsilon \mid \beta) = \Phi(q + \varepsilon h) - \Phi(q) - \varepsilon D\Phi(q)[h],$$

we get

$$\frac{\mathrm{d}^2}{\mathrm{d}\varepsilon^2}\,\mathrm{B}_\Phi(\alpha_\varepsilon \mid \beta)\bigg|_{\varepsilon=0} = D^2\Phi(q)[h,h].$$

Now compute the second variation of the Csiszár side:

$$\begin{aligned}\mathrm{D}_\varphi(\alpha_\varepsilon \mid \beta) &= \int_{\mathcal{X}} q\,\varphi\Big(\frac{q+\varepsilon h}{q}\Big)\,\mathrm{d}\lambda \\ &= \int_{\mathcal{X}} q\,\varphi\Big(1+\varepsilon\frac{h}{q}\Big)\,\mathrm{d}\lambda.\end{aligned}$$

Differentiating twice at $\varepsilon = 0$ gives

$$\frac{\mathrm{d}^2}{\mathrm{d}\varepsilon^2}\,\mathrm{D}_\varphi(\alpha_\varepsilon \mid \beta)\bigg|_{\varepsilon=0} = \varphi''(1)\int_{\mathcal{X}} \frac{h^2}{q}\,\mathrm{d}\lambda.$$

Thus

$$D^2\Phi(q)[h,h] = \varphi''(1)\int_{\mathcal{X}} \frac{h^2}{q}\,\mathrm{d}\lambda.$$

Set

$$c = \varphi''(1).$$

Since φ is convex,

$$c \geq 0.$$

The entropy functional satisfies

$$D^2H_\lambda(q)[h,h] = \int_{\mathcal{X}} \frac{h^2}{q}\,\mathrm{d}\lambda.$$

Therefore, for

$$\Psi = \Phi - cH_\lambda,$$

we have

$$D^2\Psi(q)[h,h] = 0$$

for every $q \in \mathcal{P}_{\lambda,*}$ and every bounded zero-mass direction h.

Now take arbitrary $p\lambda, q\lambda \in \mathcal{P}_{\lambda,*}$, and define the line segment

$$p_t = (1-t)q + tp, \qquad 0 \leq t \leq 1.$$

Then $p_t\lambda \in \mathcal{P}_{\lambda,*}$, and

$$\int_{\mathcal{X}} (p-q)\,\mathrm{d}\lambda = 0.$$

Hence

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\Psi(p_t) = D^2\Psi(p_t)[p-q,p-q] = 0.$$

Thus $t \mapsto \Psi(p_t)$ is affine. In particular,

$$\Psi(p) - \Psi(q) - D\Psi(q)[p-q] = 0.$$

That is,

$$\mathrm{B}_\Psi(p\lambda \mid q\lambda) = 0.$$

Since

$$\Phi = cH_\lambda + \Psi,$$

we obtain

$$\mathrm{B}_\Phi(p\lambda \mid q\lambda) = c\,\mathrm{B}_{H_\lambda}(p\lambda \mid q\lambda) = c\,\mathrm{KL}(p\lambda \mid q\lambda).$$

It remains to identify φ. Since

$$\mathrm{D}_\varphi(\alpha \mid \beta) = c\,\mathrm{KL}(\alpha \mid \beta) = \mathrm{D}_{c\,t\log t}(\alpha \mid \beta),$$

we have

$$\mathrm{D}_g(\alpha \mid \beta) = 0$$

for all $\alpha, \beta \in \mathcal{P}_{\lambda,*}$, where

$$g(t) = \varphi(t) - c\,t\log t.$$

Also,

$$g(1) = 0.$$

Choose arbitrary numbers

$$0 < x < 1 < y.$$

Set

$$\theta = \frac{y-1}{y-x}.$$

Then

$$0 < \theta < 1$$

and

$$\theta x + (1-\theta)y = 1.$$

Choose $A \in \mathcal{A}$ with

$$0 < \lambda(A) < \lambda(\mathcal{X}),$$

and put $A^c = \mathcal{X} \setminus A$. Define

$$q = \frac{\theta}{\lambda(A)}\mathbf{1}_A + \frac{1-\theta}{\lambda(A^c)}\mathbf{1}_{A^c},$$

and

$$p = x\,\frac{\theta}{\lambda(A)}\mathbf{1}_A + y\,\frac{1-\theta}{\lambda(A^c)}\mathbf{1}_{A^c}.$$

Then

$$\int_{\mathcal{X}} q\,\mathrm{d}\lambda = 1,$$

and, because $\theta x + (1-\theta)y = 1$,

$$\int_{\mathcal{X}} p\,\mathrm{d}\lambda = 1.$$

Moreover,

$$\frac{p}{q} = x \quad \text{on } A, \qquad \frac{p}{q} = y \quad \text{on } A^c.$$

Since $\mathrm{D}_g(p\lambda \mid q\lambda) = 0$, we get

$$0 = \theta g(x) + (1-\theta)g(y).$$

Substituting

$$\theta = \frac{y-1}{y-x}, \qquad 1-\theta = \frac{1-x}{y-x},$$

we obtain

$$0 = \frac{y-1}{y-x}g(x) + \frac{1-x}{y-x}g(y).$$

Multiplying by $y - x$,

$$(y-1)g(x) + (1-x)g(y) = 0.$$

Equivalently,

$$(y-1)g(x) = (x-1)g(y),$$

and hence

$$\frac{g(x)}{x-1} = \frac{g(y)}{y-1}.$$

Because $0 < x < 1 < y$ were arbitrary, the ratio

$$\frac{g(t)}{t-1}$$

is constant on $(0, \infty) \setminus \{1\}$. Thus there exists $a \in \mathbb{R}$ such that

$$g(t) = a(t-1)$$

for every $t > 0$. Therefore

$$\varphi(t) = c\, t \log t + a(t-1).$$

This proves the claim. $\square$

6 Normalizations

The generator of a Csiszár divergence is not unique: adding a linear term $a(t-1)$ does not change the divergence, because

$$\int_{\mathcal{X}} q \left(\frac{p}{q} - 1\right) \mathrm{d}\lambda = \int_{\mathcal{X}} p \,\mathrm{d}\lambda - \int_{\mathcal{X}} q \,\mathrm{d}\lambda = 0.$$

Thus

$$\varphi(t) = c\, t \log t + a(t-1)$$

and

$$\varphi(t) = c\, t \log t$$

define the same divergence on probability measures.

If one imposes the common normalization

$$\varphi'(1) = 0,$$

then

$$0 = \varphi'(1) = c + a,$$

so

$$a = -c,$$

and

$$\varphi(t) = c\big(t \log t - t + 1\big).$$

If one also imposes

$$\varphi''(1) = 1,$$

then $c = 1$, and the normalized generator is

$$\varphi(t) = t \log t - t + 1.$$

It gives exactly

$$\mathrm{D}_{\varphi}(\alpha \mid \beta) = \mathrm{KL}(\alpha \mid \beta).$$

7 Independence of the reference measure

Although H_λ uses a reference measure, its Bregman divergence does not. Indeed, suppose λ' is another equivalent reference measure, with

$$\mathrm{d}\lambda' = s\,\mathrm{d}\lambda, \qquad s > 0.$$

If $\mu = p\lambda = p'\lambda'$, then

$$p' = \frac{p}{s}.$$

Hence

$$\begin{aligned} H_{\lambda'}(\mu) &= \int_{\mathcal{X}} p' \log p' \,\mathrm{d}\lambda' \\ &= \int_{\mathcal{X}} \frac{p}{s} \log\left(\frac{p}{s}\right) s\,\mathrm{d}\lambda \\ &= \int_{\mathcal{X}} p \log p \,\mathrm{d}\lambda - \int_{\mathcal{X}} p \log s \,\mathrm{d}\lambda. \end{aligned}$$

Thus

$$H_{\lambda'}(\mu) = H_\lambda(\mu) - \int_{\mathcal{X}} \log s \,\mathrm{d}\mu.$$

The difference between $H_{\lambda'}$ and H_λ is affine in μ, so their Bregman divergences are identical. Therefore

$$\mathrm{B}_{H_\lambda}(\alpha \mid \beta) = \mathrm{KL}(\alpha \mid \beta)$$

is a measure-theoretic object, not an artifact of the choice of λ.

Remark 1. *For arbitrary probability measures, the KL divergence is understood with the usual extended convention*

$$\mathrm{KL}(\alpha \mid \beta) = \begin{cases} \displaystyle\int_{\mathcal{X}} \log\left(\frac{\mathrm{d}\alpha}{\mathrm{d}\beta}\right) \mathrm{d}\alpha, & \alpha \ll \beta, \\ +\infty, & \alpha \not\ll \beta. \end{cases}$$

For the nonzero common divergence, the corresponding Csiszár generator is

$$\varphi(t) = c\,t \log t + a(t-1), \qquad c > 0.$$

Its extended Csiszár divergence is

$$\mathrm{D}_\varphi(\alpha \mid \beta) = c\,\mathrm{KL}(\alpha \mid \beta).$$